Capgemini

APEX COMMERCIAL BANK AI LABS

✓ PRODUCTION GRADE ARCHITECTURE • V1.0.0

Enterprise AI Banking & RAG Platform

An enterprise-grade, high-security multi-role financial ecosystem fusing **Microsoft SQL Server 2022**, **FastAPI**, **Angular 21**, and the **Capgemini Generative Engine** to deliver hallucination-free document intelligence, zero-leakage natural language Text-to-SQL, and ultra-low-latency query caching.

- Multi-Format Ingestion (PDF / JSON / TXT)
- Strict 3-Role RBAC (Customer • Staff • Admin)
- User-Scoped Text-to-SQL with PII Masking
- Multi-Model LLM Gateway (GPT-5, GPT-4o, Claude 3.5, Grok-4)
- Sub-5ms Query Cache

~5 ms	1536-D	100%	13 Tables
CACHED QUERY LATENCY	VECTOR DIMENSION	PII MASKED & ISOLATED	MS SQL RELATIONAL CORE
0 Token Cost via SHA-256 Cache	text-embedding-3-small Cosine Match	AC*******1023 • 4111********1234	ACID Compliant + Audit Logging

5 Core System Use Cases & Architectural Solutions

1. Foundation & Multi-Format Vector Retrieval DATA FOUNDATION

Automated ingestion pipeline supporting **PDF**, structured **JSON**, and **TXT**. Splits text at **800-token boundaries** with **120-token window overlap** (`cl100k_base` BPE). Generates 1536-dimensional embeddings indexed inside Microsoft SQL Server for exact cosine similarity lookup.

`pypdf` `tiktoken (cl100k_base)` `text-embedding-3-small` `MS SQL Vector Store`

2. End-to-End RAG Banking Assistant with Citations ZERO HALLUCINATION

Real-time policy assistant answering compliance, loan guidelines, and deposit penalty inquiries. Enforces strict system prompts to eliminate hallucinations and attaches a **Source Citations Engine** linking every response back to exact filename, document title, chunk index, and cosine relevance score.

`Top-K Retrieval` `Strict System Prompting` `Source Citations Engine`

3. Secure Banking Data Integration (Text-to-SQL) SECURITY FIRST

Customers query balances, credit cards, loan repayment schedules, and transaction ledgers in natural language. Enforces absolute security: strictly constrained by `WHERE user_id = :current_user_id`, parameterized read-only execution, and automated PII redaction (`AC*******1023`, `4111********1234`).



Multi-Model LLM Gateway:

- `openai.gpt-5`
- `openai.gpt-4o`
- `claude-opus-4-6`
- `grok-4`

End-to-End Request & Intelligence Flow

Angular 21 Client	FastAPI Gateway	Guardrails & PII
Port 4200 · Signals		Masking Engine

User-Scoped SQL | Anti-Injection Scanner | Automated PII Masking

4. Intelligence Maturity & Query Caching

5MS OPTIMIZATION

High-efficiency **Conversational Query Caching** using normalized SHA-256 hashes stored in MS SQL. Frequent and identical policy questions resolve in **~5ms with 0 token consumption**. Full computational telemetry (`/metrics`) tracks latency, token burn, and cache hit ratios.

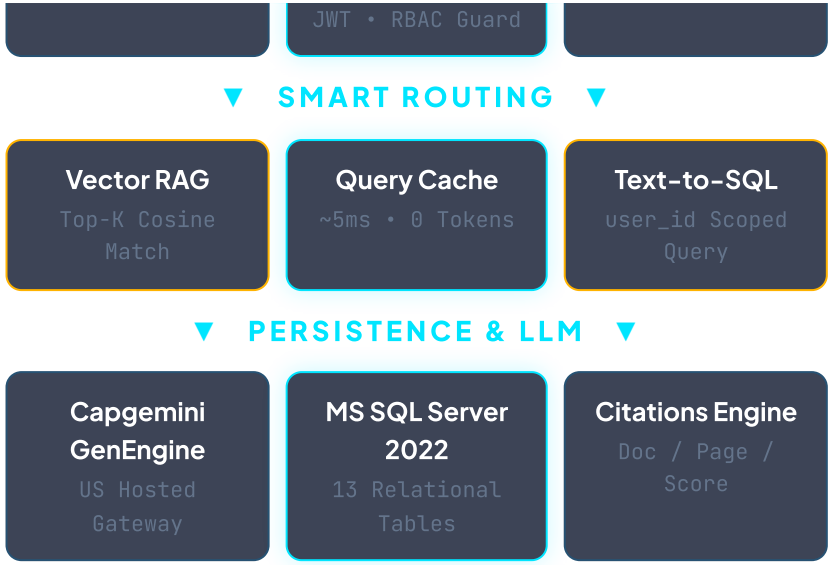
SHA-256 Query Hashing | Zero Token Cost | Real-time /metrics Telemetry

5. Unified Integration & Smart Intent Classifier

GATEWAY INTELLIGENCE

Centralized API gateway router that parses natural language intent dynamically into `POLICY_DOC` (Vector RAG), `USER_SQL` (Personal Database), or `REFUSE_OOS` (Out-of-Scope Security Refusal), delivering a unified single-portal experience across all roles.

Intent Classifier | RBAC Token Validator | Dynamic Route Dispatch



Enterprise 3-Role Governance & Segregation of Duties

CUSTOMER (Retail Banking)

- Self-registration assigns `CUSTOMER` & `KYC_PENDING`

EMPLOYEE (Bank Staff Operations)

- Exclusively provisioned by Admin (zero public registration)

ADMIN (System Super-User)

- Auto-seeded root account (`admin@capgemini-bank.com`)

- Interactive Dashboard: Balances, credit limits, loans & ledger
- Document verification upload (PAN, Aadhaar, Passport)
- Natural language AI assistant for policy & scoped account insights
- Strict `403 Forbidden` block against staff queues & admin APIs

- Pending KYC Workspace: Audits identity documents
- Core Account Activation: Issues unique 10-digit account code
- Initial credit provisioning & Platinum Rewards card issuance
- Customer ticket review & strict administrative isolation

- Employee provisioning & organizational credential management
- Multi-format knowledge ingestion pipeline (PDF, JSON, TXT)
- Telemetry dashboard: Latency benchmarks, token usage, hit ratio
- Compliance audit logs inspection with immutable event tracking

🛠️ Core Technology Stack Specifications

BACKEND LAYER	DATABASE ENGINE	FRONTEND FRAMEWORK	AI PLATFORM	ENTERPRISE SECURITY
FastAPI • Python 3.12	**MS SQL Server 2022**	**Angular 21**	**Capgemini GenEngine**	**JWT • BCrypt • PII**
Asynchronous ASGI, Pydantic v2, SQLAlchemy 2.0	ODBC Driver 18, 13 Relational Tables, pyodbc pool	Standalone Components, Reactive Signals, TypeScript	GPT-5, GPT-4o, Claude Opus, Grok-4, text-embedding-3	24h Tokens, Salted Hashing, Account & Card Redaction